

RÉSUMÉ

Michael A. Boateng

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EDUCATION AND QUALIFICATIONS

School of Engineering, Georgia Institute of Technology, Atlanta, GA

Aug 2023 – Jun 2028

Doctor of Philosophy (Ph.D.), Electrical and Computer Engineering.

- Graduate Research Assistant – AI Institute for Advances in Optimization (AI4OPT) | Energy Division
- **Georgia Tech Supervisors:** Prof. Pascal Van Hentenryck and Prof. Daniel Molzahn
- **External Industry and Academic Mentors:** Dr. Russell Bent (LANL) and Prof. Parikshit Pareek (IIT Roorkee)

School of Engineering, Georgia Institute of Technology, Atlanta, GA

Aug 2023 – May 2025

Master of Science (M.Sc.), Electrical and Computer Engineering. CGPA: 3.80/4.0

- Minor: Industrial and Systems Engineering (ISyE) – *Regression Analysis, Convex Optimization & Machine Learning*
- Coursework: **Power Systems** (*Protection, Cybersecurity, Operation & Control*), **Linear Systems & Control**

Engineering Department, Ashesi University, Accra, Ghana

Sep 2018 – Jun 2022

Bachelor of Science (B.Sc.), Electrical and Electronics Engineering. Cum laude

- B.Sc. Project: Overall Best Thesis – Engineering Department [Published in IEEE Xplore | Citations: 10]

RESEARCH EXPERIENCE

Graduate Research Consultant, Austin, TX

Oct 2025 – Present

Electric Reliability Council of Texas (ERCOT), *Risk Assessment of Texas Grid Using AC SCOPF*

- Collaborating with ERCOT engineers on machine learning approaches for Security-Constrained AC Optimal Power Flow (AC SCOPF) and greater-than N-4 contingency analysis, using real-world operational data from the Texas grid
- Developing a self-supervised primal-dual learning neural-network that incorporates AC non-linearities into preventive SCOPF pipelines, with the goal of enhancing grid reliability under large-scale system conditions (1000+ buses)

Graduate Research Intern, Los Alamos, NM

May 2025 – Present

Los Alamos National Laboratory (LANL), *End-to-End Learning for AC Feasibility Restoration in DCOPF*

- Designing an LMP-preserving DCOPF-ACPF pipeline that learns DCOPF parameters and generator participation factors, and computes pseudo-LMPs from AC-PF Jacobian sensitivities to keep AC prices aligned with DC outcomes
- Implementing implicit-function gradients—KKT adjoints for DC-OPF and a single transpose-Jacobian solve for AC-PF—to enable end-to-end training in Python with NumPy/SciPy, PyTorch, CVXPY, and OSQP/Gurobi
- Prototyping fast feasibility restoration with smooth voltage-reactive power handling, while exploring tap/shunt policies to cut recalculations and runtime on large test cases by 90%, compared to conventional approaches using LANL clusters

Graduate Research Intern, Los Alamos, NM

Sept 2024 – Aug 2025

Los Alamos National Laboratory (LANL), *Exploring DCOPF and ACPF Formulations for AC Feasibility*

- Engineered a unified DCOPF→ACPF recovery pipeline combining four loss-aware DC models with AC feasibility restoration techniques including distributed slack allocation, PV/PQ bus switching, and voltage-informed initialization
- Evaluated the pipeline on 10,000+ stochastic dispatch scenarios across large-scale power systems, demonstrating feasibility restoration in the major cases and reducing inequality constraint violations by 3–5× under extreme loading
- Reduced Newton-Raphson convergence iterations by ~50% in stressed test cases and eliminated active/reactive power violations across challenging industry-grade networks, including PEGASE and RTE benchmark systems
- Achieved 75% lower mean absolute error and 93% cost difference reduction compared to traditional single-slack methods, showing the effectiveness of incorporating line losses and distributed control in DCOPF formulations

Graduate Research Assistant, Atlanta, GA

Apr 2025 – Oct 2025

AI Institute for Advances in Optimization (AI4OPT), *Volt-VAR Optimization for Large-Scale Power Systems*

- Designed a transmission-scale Volt/VAR Optimization framework that co-optimizes OLTC tap ratios and capacitor-bank modules under AC power-flow physics, implementing a relax-round-resolve pipeline, that encodes device limits
- Formulated operations-aligned objectives that hold voltage and VAR setpoints while penalizing active-power redispatch and preserving economic dispatch, integrating ACOPF/market references for AC-feasible setpoints
- Performed an extensive review of transmission-scale Volt/VAR and discrete ACOPF (MINLP, relax-round-resolve, cutting-plane/OA, homotopy, metaheuristics, interior-point); pinpointing gaps in (i) scalable AC-feasible tap/shunt schedules, (ii) realistic device step modeling, and (iii) coordination with market dispatch and loss/price sensitivities

Graduate Research Assistant, Atlanta, GA

May 2025 – Present

Georgia Institute of Technology (Georgia Tech), *Neural Network-Accelerated Dynamic State Estimation for CT Error*

- Benchmarked 9 Dynamic State Estimation (DSE) algorithms combining iterative solvers (Gauss-Newton, Levenberg-Marquardt, Gradient Descent) with integration types (Euler, Trapezoidal, Quadratic) on 400 COMTRADE samples
- Formulated CT saturation correction as unconstrained Weighted Least Squares optimization embedding nonlinear circuit dynamics at 10^{-4} tolerance, establishing baseline metrics for real-time merging unit deployment
- Designing PINNs with unrolled Gauss-Newton layers to accelerate CT error correction to 5-10 iterations per sample, and incorporating learnable damping parameters to achieve increased speedups and accuracy on *GTech computer clusters*

Graduate Research and Teaching Assistant, Atlanta, GA

Oct 2024 – Present

Georgia Institute of Technology (Georgia Tech), *Design Fundamentals (ECE 3011)*, *Digital Design Lab (ECE 2031)*

- Introduced faculty and students to Electronic Design Automation (EDA) as an alternative to Eagle and Altium, and created a full set of lecture slides and video guides to support final-project Printed Circuit Board (PCB) design
- Leading the development of MPower Boards (MPBs), integrating 3-D printing, LED technology, and real-time simulations to create a physical, interactive model of power grids for enhanced educational and research
- Advancing commercialization with Georgia Tech's Create-X program by building a portable, battery-powered device, scoping IP for a provisional patent, and piloting IEEE-14/30/39 classroom demos, to visualize grid line faults

SKILLS

- **Languages:** English (Native), Sign Language, C++, MATLAB, PHP, MySQL, HTML, CSS, R, Python, and Julia
- **Tools:** PowerModels, MATPOWER, PG-LIB, Pandapower, PowerWorld, PyPSA, SciPy, CVXPY, PyTorch, Git

HONORS AND AWARDS

LANL Grid Science Winter School & Conference: Travel Grant Winner, NM, USA	Jan 2025
Argonne Department of Energy's CyberForce® Competition: Travel Grant Winner, IL, USA	Nov 2024
International Solid-State Circuits Conference: Travel Grant Winner (1 of 4, 1 st Year ECE PhDs), CA USA	Jan 2024
National Society of Black Engineers Conference: NSBE50 Scholarship, GA USA	Mar 2024
Silver Medalist: Poster Presentation, International Genetically Engineered Machine (<i>Competition</i>), France	Oct 2022
1 st Place: Engineering Poster Presentation, Statistics Poster Session (<i>College</i>), Ghana	Dec 2019
1 st Place: Innovation Research Competition, Royal Crown Packaging Limited (<i>Company</i>), Ghana	Jun 2019

RECENT PRE-PRINTS/ PUBLICATIONS

- **M.A. Boateng**, R. Bent, S. Misra, P. Park, P. Van Hentenryck, and D.K. Molzahn, "Towards AC Feasibility of DCOF Dispatch," *Submitted* to the 24th Power Systems Computation Conference (PSCC), 2025.
- S. Tong, **M.A. Boateng**, M. Tanneau, and P. Van Hentenryck, "Volt/VAR Optimization in Transmission Networks with Discrete-Control Devices," *Submitted* to the 24th Power Systems Computation Conference (PSCC), 2025.
- **M.A. Boateng**, G. Gauderman, and B. Ashebo, "Dynamic State Estimation in Current Transformer Merging Units: Evaluating Iterative and Integration Methods for Enhanced Efficiency," *Abstract Submitted* to the 2026 Georgia Tech Protective Relaying Conference (GTPRC), 2025.
- Asiamah, R., Talkington, S., **Boateng, M.**, Vanin, M., Geth, F., & Molzahn, D. (2025). "Classifying Reactive Power Control Laws of Behind-the-Meter Solar Photovoltaic Inverters." *In Proceedings* of the IEEE Kansas Power and Energy Conference (KPEC).

PRESENTATIONS/ TALKS

AI Institute for Advances in Optimization, Industry Partners Meeting, GA, USA	Nov 2025
Los Alamos National Lab, Lightning Talks T-5 Division, NM, USA	Jul 2025
IEEE International Conference on Engineering and Emerging Technologies, Kuala Lumpur, Malaysia	Oct 2022

REVIEWED CONFERENCE/ JOURNAL PAPERS

- 2025 24th *Power Systems Computation Conference (PSCC)*, Track: AC-Power Flow, and Volt-Var Optimization
- 2025 *Electric Power Systems Research Journal* (Elsevier, Impact Factor: 4.2; CiteScore: 8.2). Track: Neural Networks
- 2025 IEEE *Kansas Power and Energy Conference (KPEC)*, Track: Solar Power, Transmission and Distribution Systems

LEADERSHIP, ACTIVITIES, AND COMMUNITY INVOLVEMENT

President: Fellowship of Christian Graduate Students , GA, USA	Jul 2025 – Present
Vice President: Georgia Tech Ghanaian Students Association , GA, USA	Jun 2025 – Present
Chair, Special Projects and Programs: African Graduate Students Committee , GA, USA	May 2025 – Present
Co-Captain: International Genetically Engineered Machine Competition (iGEM) , Paris, France	Jun – Oct 2022
Editor-in-Chief: Ashesi Science Engineering Entrepreneurship Design (SEED) Journal , Ghana	2022 – 2023